

Replay Detection - Results

Datasets

- ASVspoof databases: <https://www.asvspoof.org/database>
- ASVspoof 2017 V2: <https://datashare.ed.ac.uk/items/59543650-e9b0-415d-8058-0567f908ce37>
- ASVspoof 2019 (PA + LA) : <https://www.kaggle.com/datasets/awsaf49/asvspoof-2019-dataset>
- Metric: EER (%)

Results

1. ASVspoof 2017 (in-domain)

Front-end	Train	Test	EER (%)
Mel	2017	2017 held-out	10.0
Inverted-Mel	2017	2017 held-out	4.9

2. ASVspoof 2019 PA (inverted-Mel from scratch)

Train	Test	EER (%)
PA subset	PA dev	11.3
PA fuller train	PA dev	8.0

3. Cross-domain (zero-shot)

Train	Test	EER (%)
2017 inverted-Mel	PA2019	~50
2019 PA inverted-Mel	2017	~41.1

4. Mixed inverted-Mel (2017 + PA) – (we are using this model in the application now)

Test	EER (%)
ASVspoof 2017 dev	9.2
PA2019 dev (readable)	11.0

5. Front-end compare (mixed 2017+PA train)

Front-end	2017 EER (%)	PA EER (%)
Mel	21.2	6.0
Inverted-Mel	10.1	9.7
LFCC	9.2	9.0

** Although LFCC model has better results, it didn't perform well in real world application.

ECAPA-TDNN + WebRTC model - Results

All EER values are percentages (%).

Paper used: <https://arxiv.org/abs/2508.18913v1>

Noise augmentation was done via MUSAN dataset.

Paper didn't specify results for clean audio.

For my model:

- without model: 0.62%
- with model: 2.88%

Type	SNR (dB)	Model from the paper	Our System (ECAPA-TDNN + WebRTC)	Improvement
Noise				
Noise	0	9.43	5.37	4.06
Noise	-5	10.76	7.67	3.09
Noise	-10	14.74	11.28	3.46
Noise	-15	21.48	15.00	6.48
Noise	-20	29.90	19.90	10.00
Music				
Music	0	12.19	5.71	6.48
Music	-5	16.37	9.39	6.98
Music	-10	24.49	14.51	9.98
Music	-15	34.96	20.54	14.42
Music	-20	43.88	26.69	17.19
Babble				
Babble	0	24.04	10.21	13.83
Babble	-5	31.43	17.70	13.73
Babble	-10	38.47	26.07	12.40
Babble	-15	42.18	34.07	8.11
Babble	-20	45.38	39.63	5.75